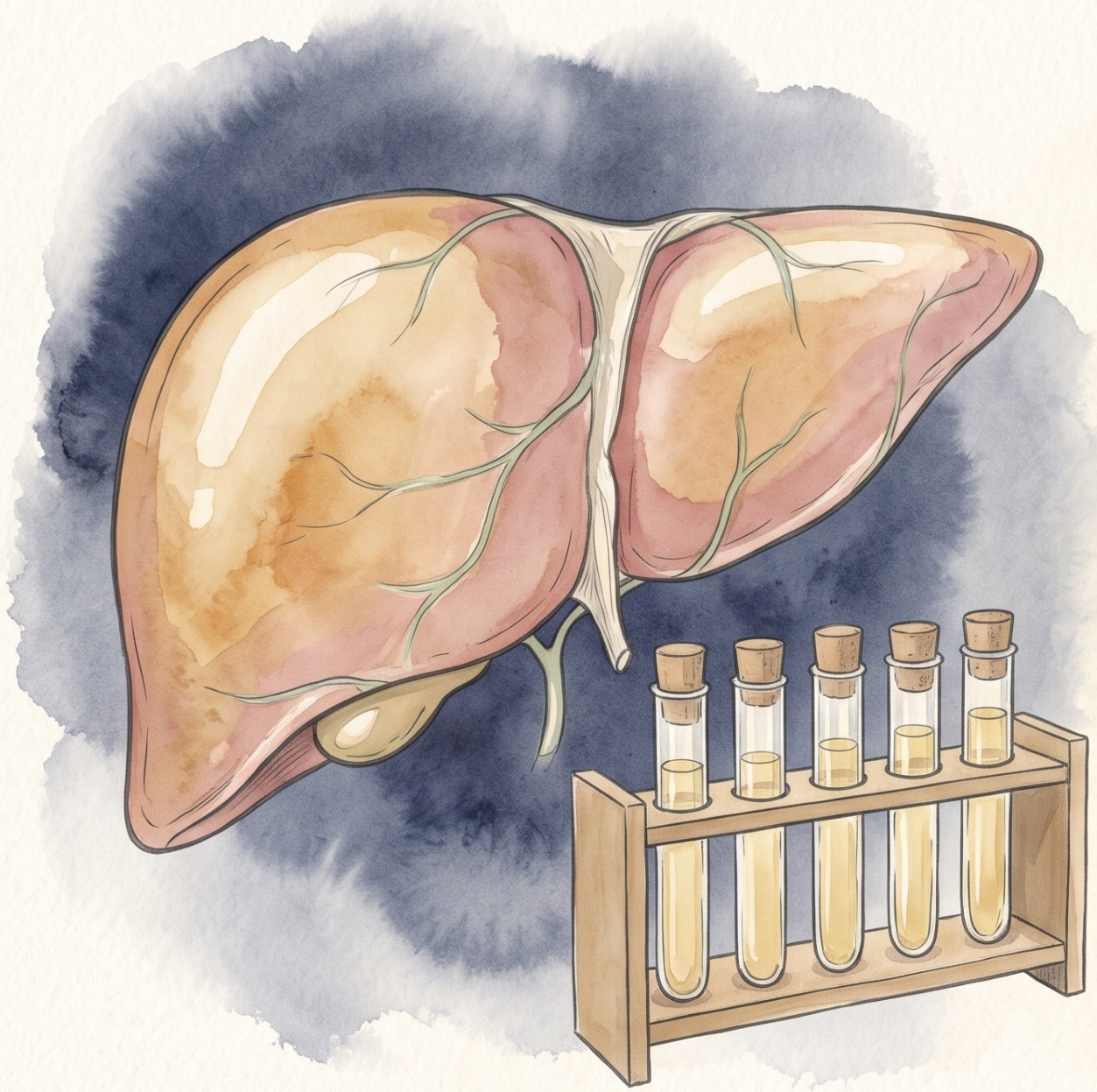


Chronic Hepatitis B

Mohammad Almeqdadi, MD
Transplant Hepatologist and Gastroenterologist
Assistant Professor of Medicine and Surgery
X: @mfalmeqdadi

Tufts UNIVERSITY | School of Medicine

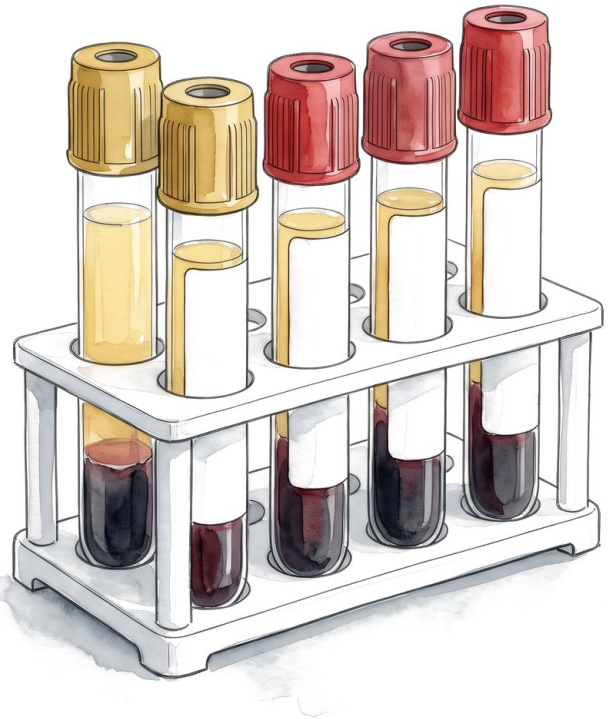
Slides (PDF)

Illustrations generated with Google Gemini

Objectives

By the end of this talk you should be able to:

- Read a hepatitis B serology panel
- Place a patient in the right phase
- Decide who to treat, using the current guidelines
- Estimate HCC risk and set up surveillance
- Prevent reactivation and transmission to the baby



The Australia antigen

A "New" Antigen in Leukemia Sera

The "Australia antigen" is found in the sera of some normal individuals from foreign populations. The total absence of the antigen from the sera of normal United States subjects and its relatively high frequency in acute leukemia suggests that the presence of the antigen may be of value in the diagnosis of early acute leukemia. Whether the antigen results from or precedes the leukemia process remains to be seen.

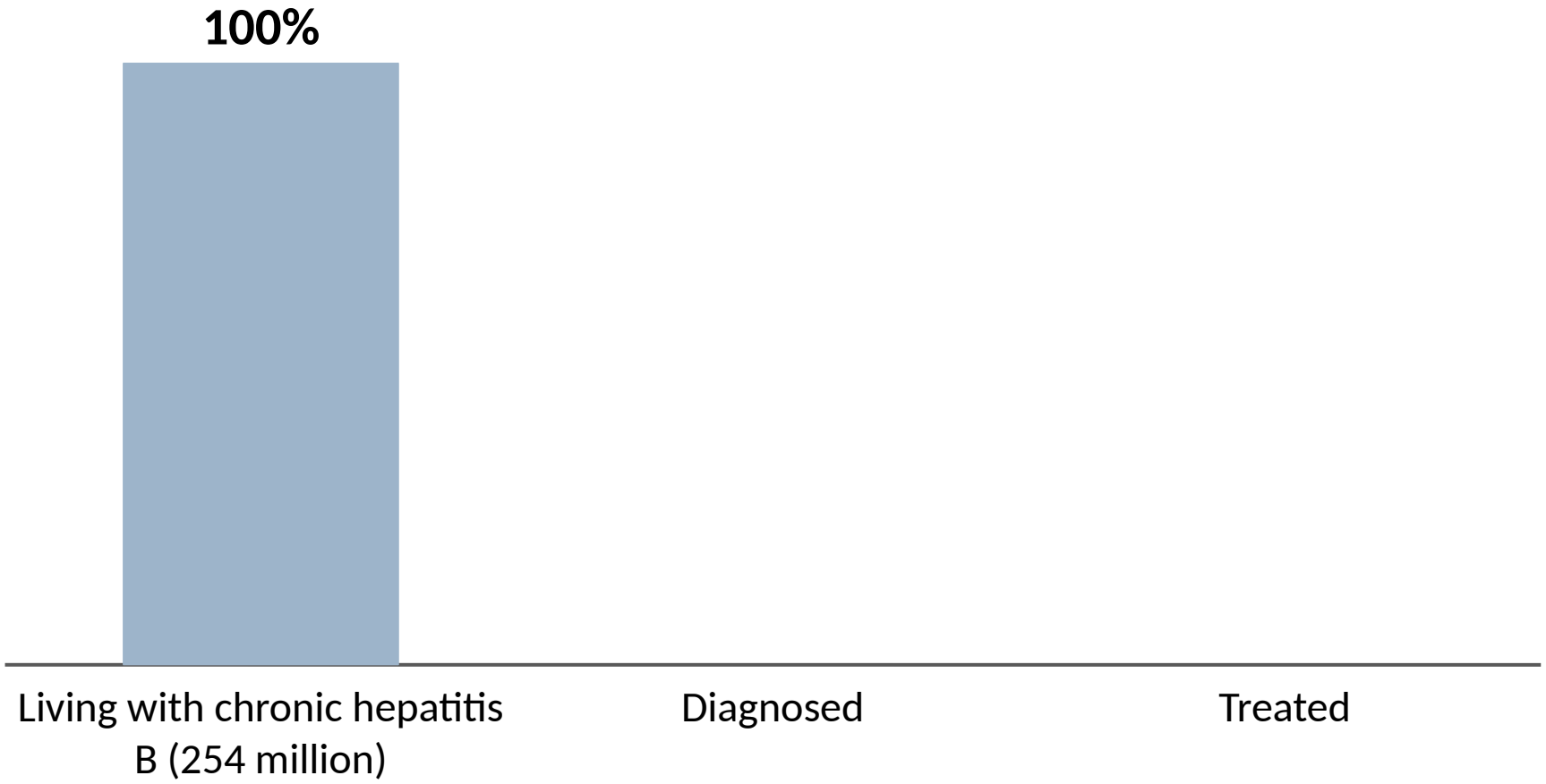
In 1963, sera from patients with hemophilia who had received transfusions were tested for the presence of isoprecipitins using a panel of 24 sera from normal individuals, including sera from foreign populations. Two of the hemophilia sera formed a clearly defined precipitin line with one of the panel sera (from an Australian aborigine), but with none

- Found in the serum of some normal people from foreign populations, and in none of the normal US subjects
- Frequent in acute leukemia, which is why the paper is about leukemia
- Blumberg shared the 1976 Nobel Prize with Carleton Gajdusek

Blumberg BS, Alter HJ, Visnich S. JAMA 1965;191:541-546. PMID 14239025.

Hepatitis B worldwide

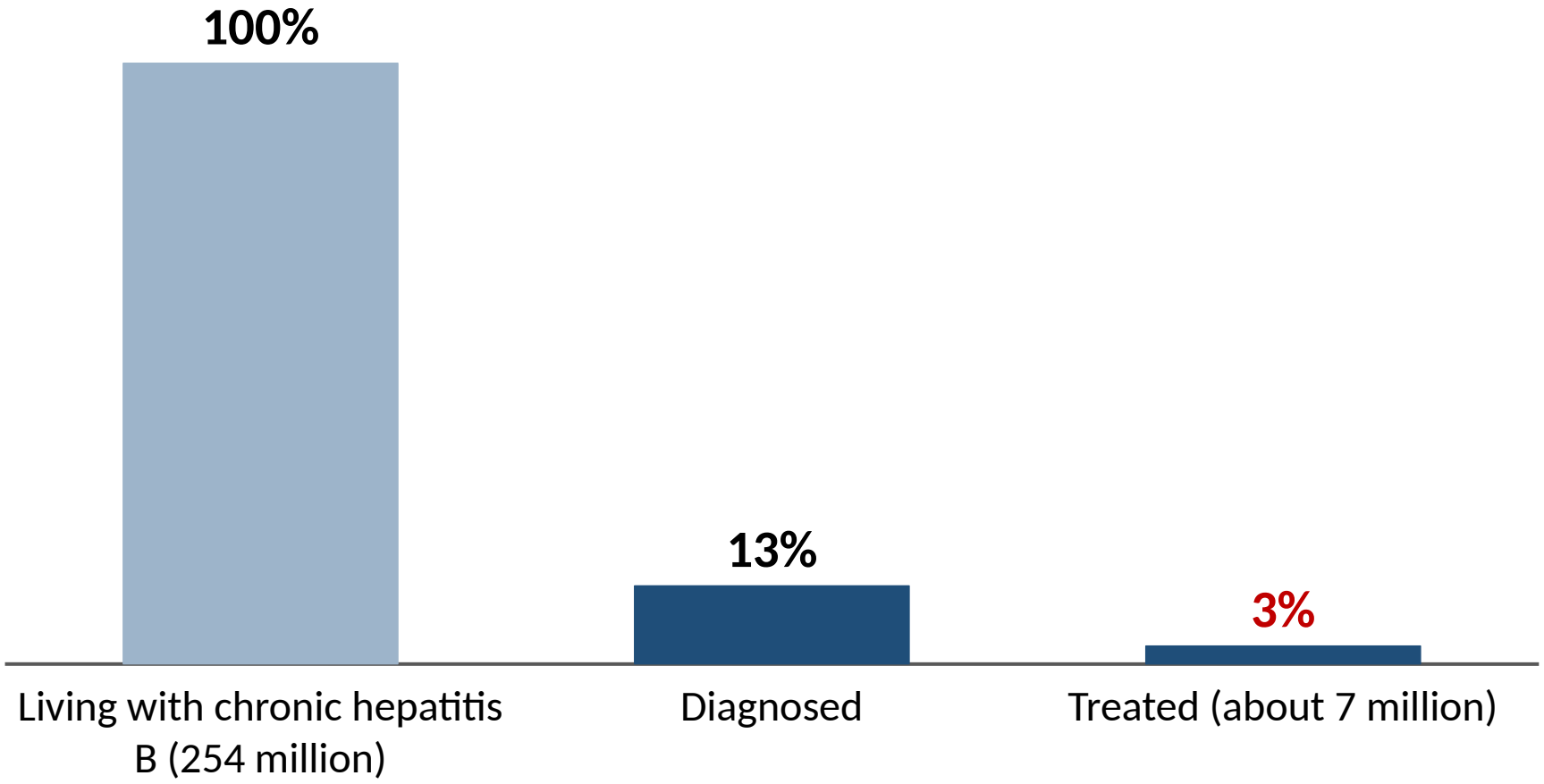
How many people with chronic hepatitis B know it?



WHO news release 9 April 2024 (2022 estimates), rounded. WHO Global hepatitis report 2024.

Hepatitis B worldwide

How many people with chronic hepatitis B know it?



WHO news release 9 April 2024 (2022 estimates), rounded. WHO Global hepatitis report 2024.

Screening an adult for hepatitis B

Which three tests make up the screening panel?

Question	Test
Is there a current infection?	
Is the patient immune?	
Has the patient ever been exposed?	

CDC recommendations, MMWR Recomm Rep 2023;72(1):1-25, PMID 36893044. Anti-HDV: EASL 2025, PMID 40348683.

Screening an adult for hepatitis B

Which three tests make up the screening panel?

Question	Test
Is there a current infection?	HBsAg
Is the patient immune?	Anti-HBs
Has the patient ever been exposed?	Total anti-HBc

Once for every adult 18 and older, and in every pregnancy

HBsAg positive: test anti-HDV once

CDC recommendations, MMWR Recomm Rep 2023;72(1):1-25, PMID 36893044. Anti-HDV: EASL 2025, PMID 40348683.

Five panels

What does each panel mean?

	HBsAg	Total anti-HBc	IgM anti-HBc	Anti-HBs	Interpretation
Panel 1	–	–	–	–	
Panel 2	–	–	–	+	
Panel 3	–	+	–	+	
Panel 4	+	+	–	–	
Panel 5	+	+	+	–	

Interpretation per CDC 2023 (Table 1), MMWR Recomm Rep 2023;72(1):1-25.

Five panels

What does each panel mean?

	HBsAg	Total anti-HBc	IgM anti-HBc	Anti-HBs	Interpretation
Panel 1	–	–	–	–	Susceptible
Panel 2	–	–	–	+	Vaccinated
Panel 3	–	+	–	+	Resolved infection
Panel 4	+	+	–	–	Chronic infection
Panel 5	+	+	+	–	Acute infection

Interpretation per CDC 2023 (Table 1), MMWR Recomm Rep 2023;72(1):1-25.

61-year-old woman starting rituximab

HBsAg	Total anti-HBc	Anti-HBs	HBV DNA	ALT
–	+	–	Not done	28 U/L

What could this be, and what do you order?

Possibility	How you sort it out

CDC 2023 (Table 1), PMID 36893044; AASLD 2018 hepatitis B guidance, PMID 29405329.

61-year-old woman starting rituximab

HBsAg	Total anti-HBc	Anti-HBs	HBV DNA	ALT
–	+	–	Not done	28 U/L

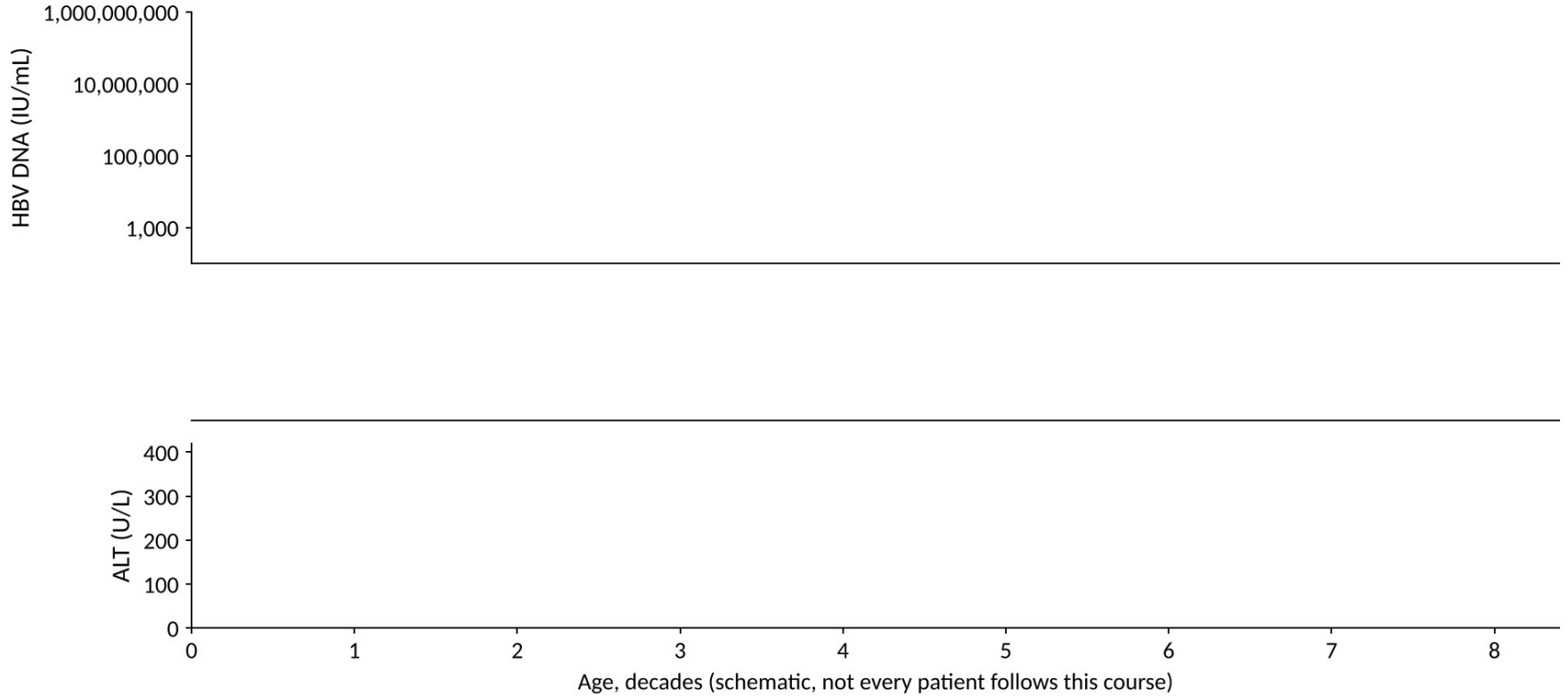
What could this be, and what do you order?

Possibility	How you sort it out
Resolved infection, anti-HBs has waned	HBV DNA undetectable
Occult infection	HBV DNA detectable
Window period	IgM anti-HBc positive
False-positive anti-HBc	Repeat testing, DNA undetectable

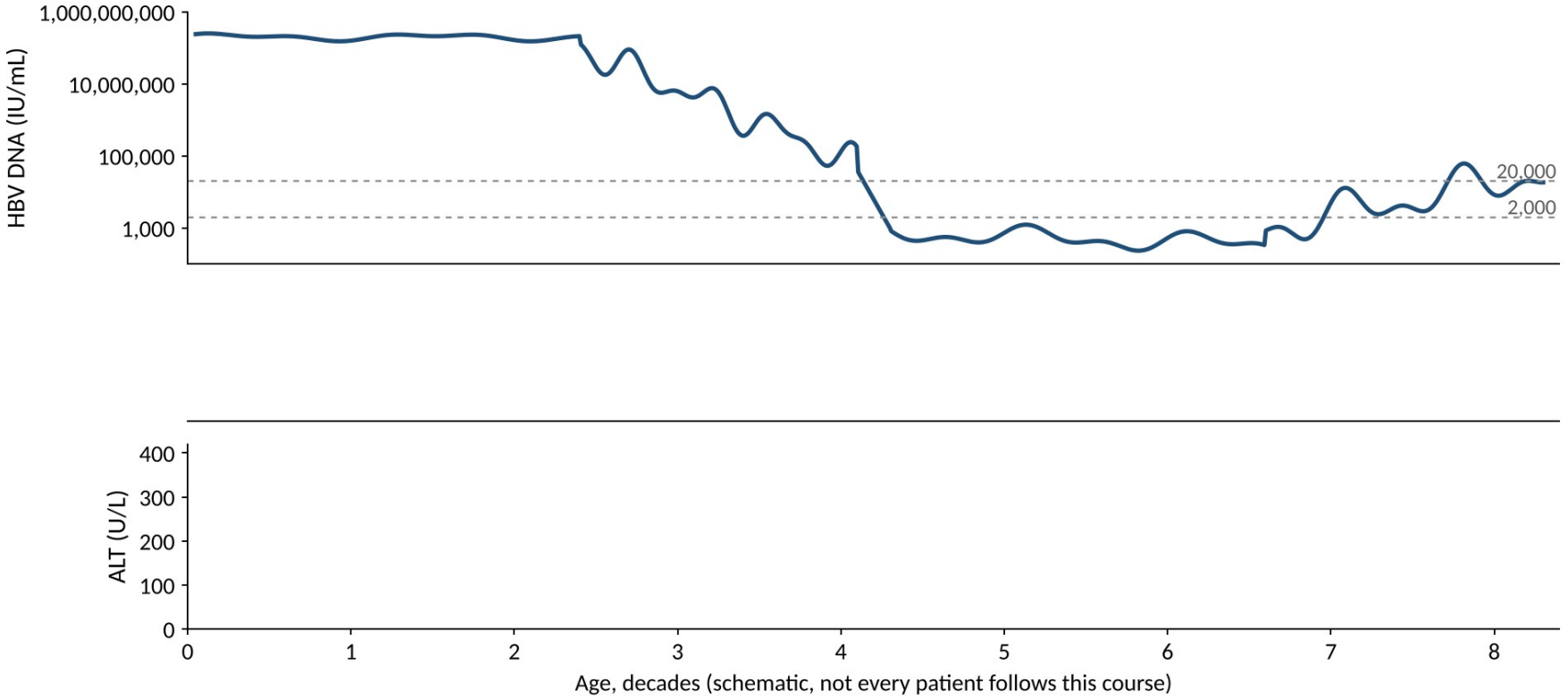
CDC 2023 (Table 1), PMID 36893044; AASLD 2018 hepatitis B guidance, PMID 29405329.

Natural history of chronic hepatitis B

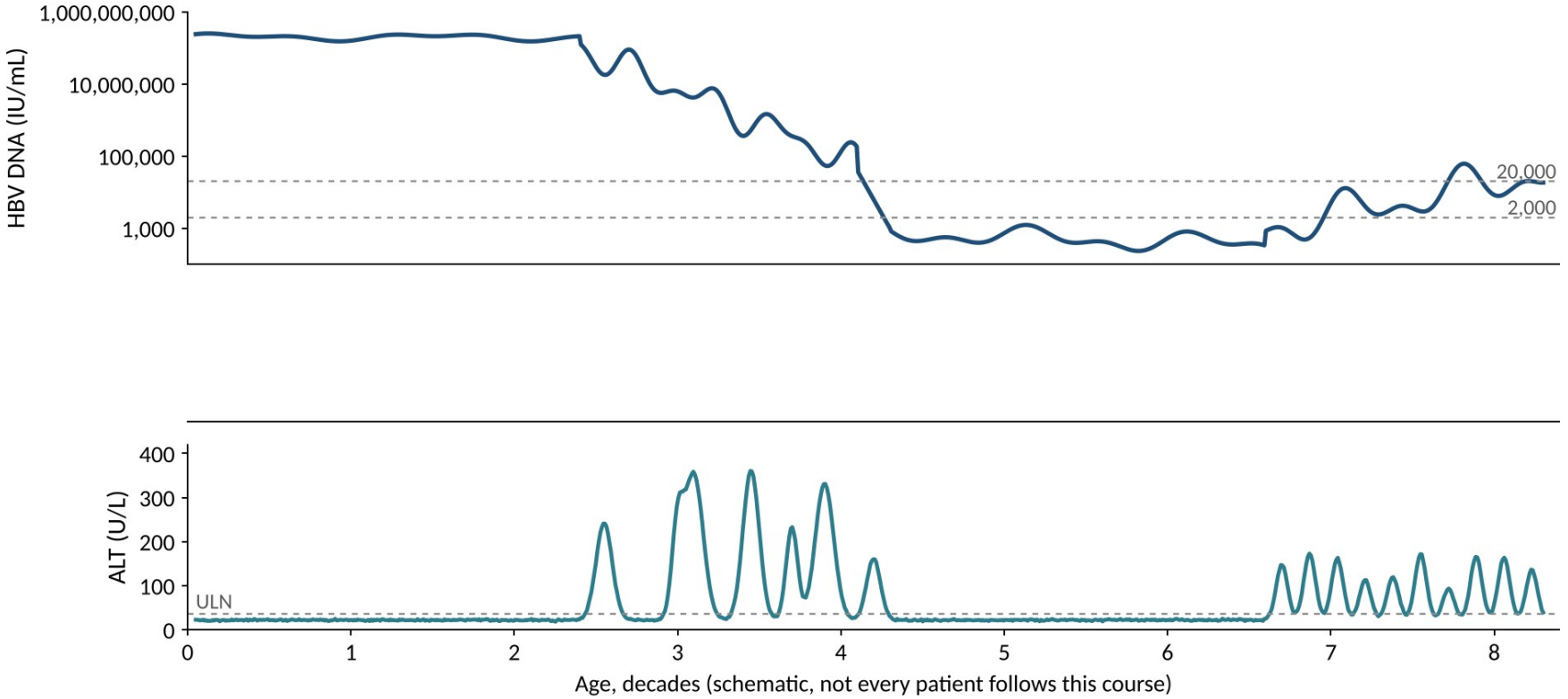
Where do HBV DNA, ALT and the antigens go over a lifetime?



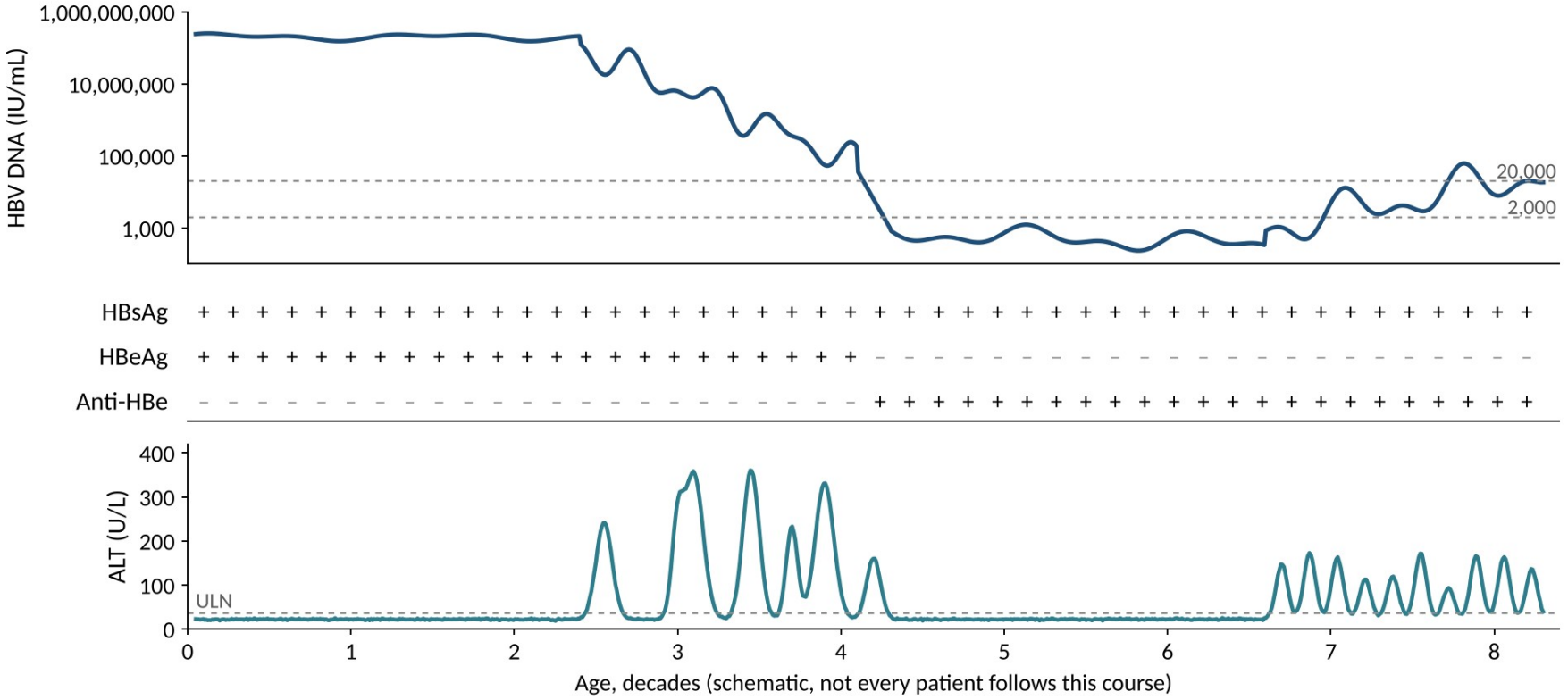
Natural history of chronic hepatitis B



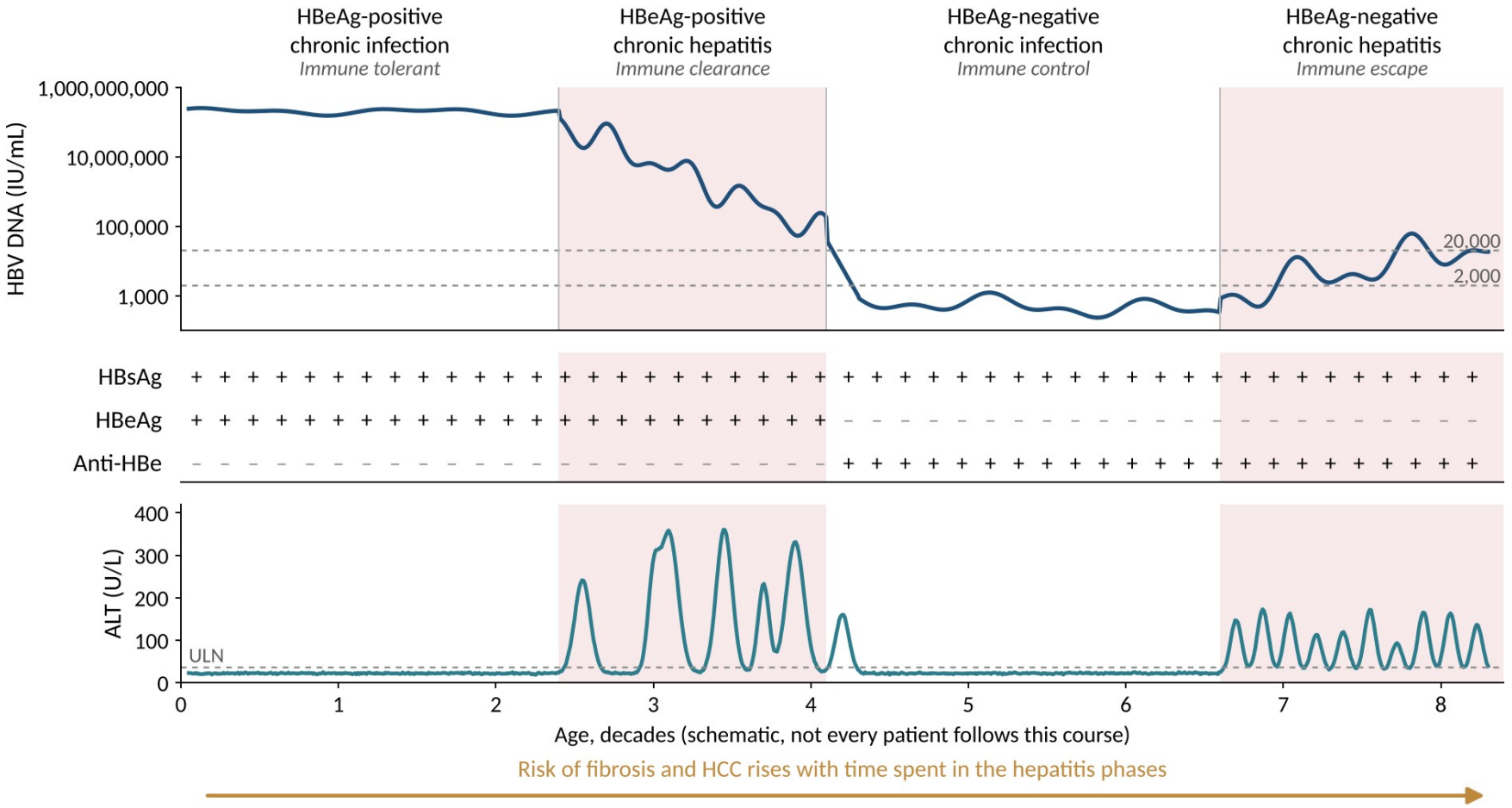
Natural history of chronic hepatitis B



Natural history of chronic hepatitis B

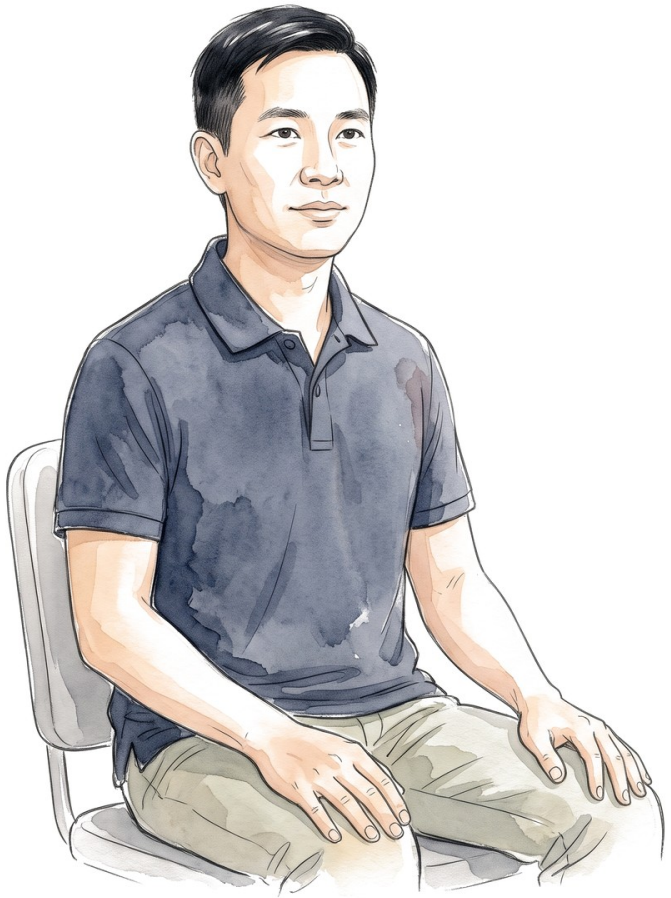


Natural history of chronic hepatitis B



34-year-old man, born in Vietnam

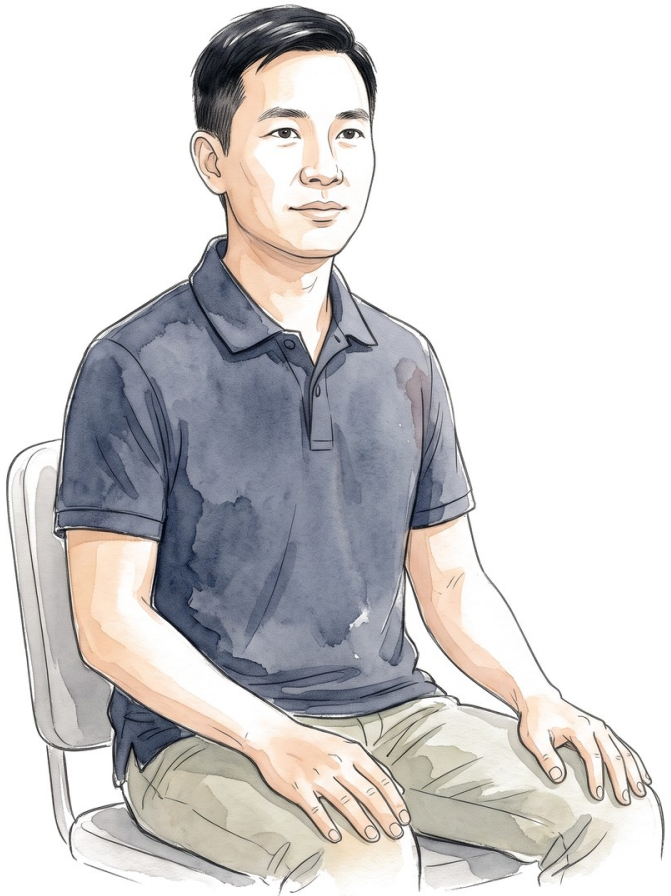
Test	Result
HBsAg	Positive
HBeAg	Positive
Anti-HBe	Negative
HBV DNA	60,000,000 IU/mL
ALT (ULN 35)	24 U/L
Platelets	245 ×10⁹/L
Liver stiffness	5.1 kPa



Do you treat him?

34-year-old man, born in Vietnam

Test	Result
HBsAg	Positive
HBeAg	Positive
Anti-HBe	Negative
HBV DNA	60,000,000 IU/mL
ALT (ULN 35)	24 U/L
Platelets	245 ×10⁹/L
Liver stiffness	5.1 kPa

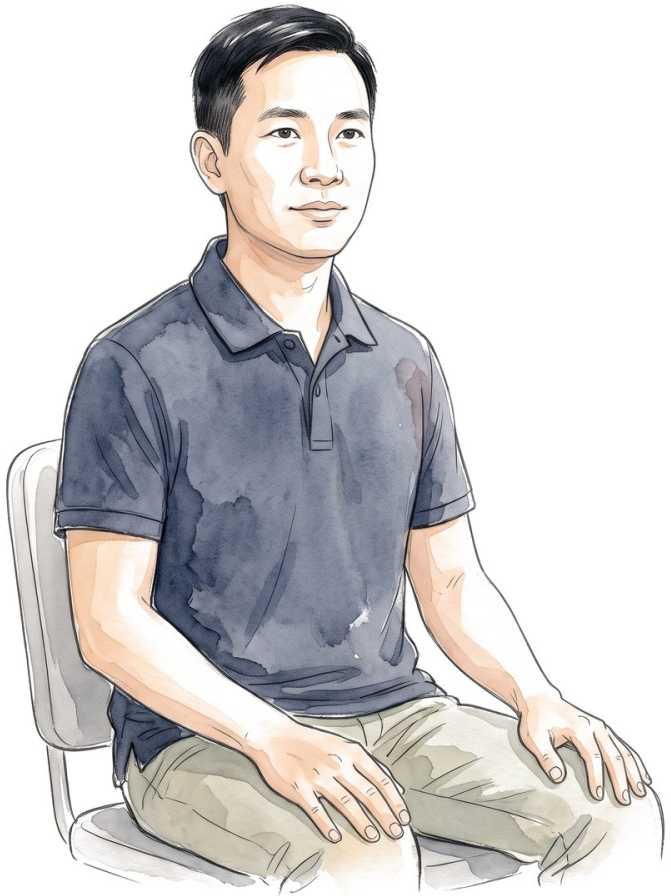


Do you treat him?

AASLD: no, monitor every 3 to 6 months. EASL 2025 would consider treatment over 30

Same man, now 41 years old

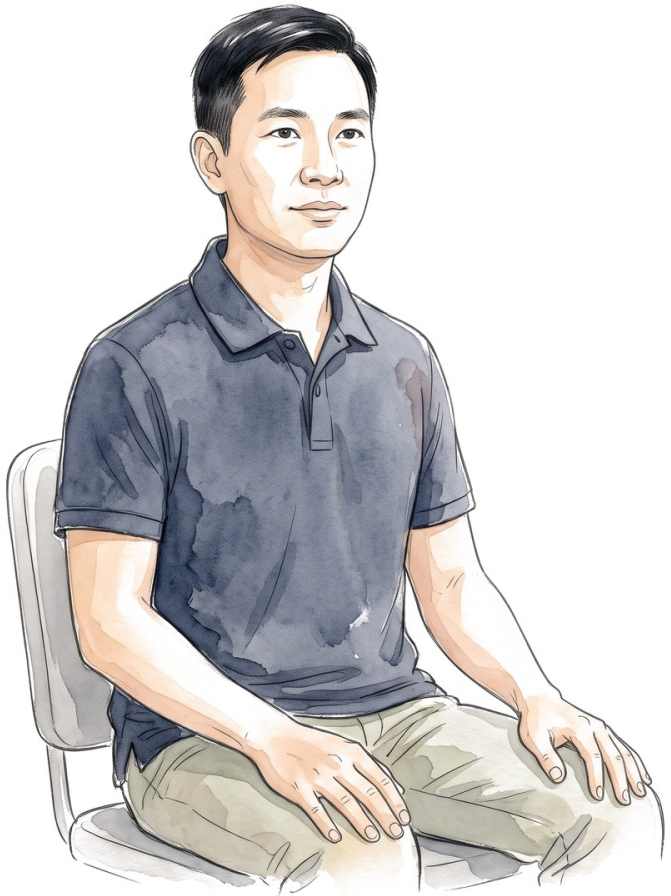
Test	Result
HBsAg	Positive
HBeAg	Positive
Anti-HBe	Negative
HBV DNA	55,000,000 IU/mL
ALT (ULN 35)	26 U/L
Platelets	245 ×10⁹/L
Liver stiffness	5.1 kPa



Does the answer change?

Same man, now 41 years old

Test	Result
HBsAg	Positive
HBeAg	Positive
Anti-HBe	Negative
HBV DNA	55,000,000 IU/mL
ALT (ULN 35)	26 U/L
Platelets	245 ×10⁹/L
Liver stiffness	5.1 kPa

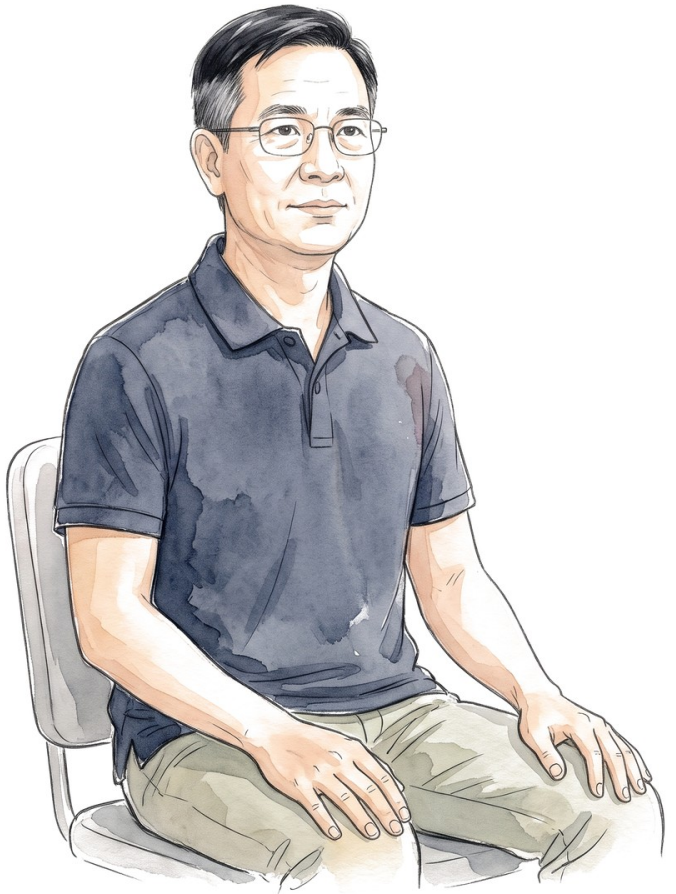


Does the answer change?

Over 40: treatment is suggested. He chose to wait.
Start ultrasound and AFP every 6 months

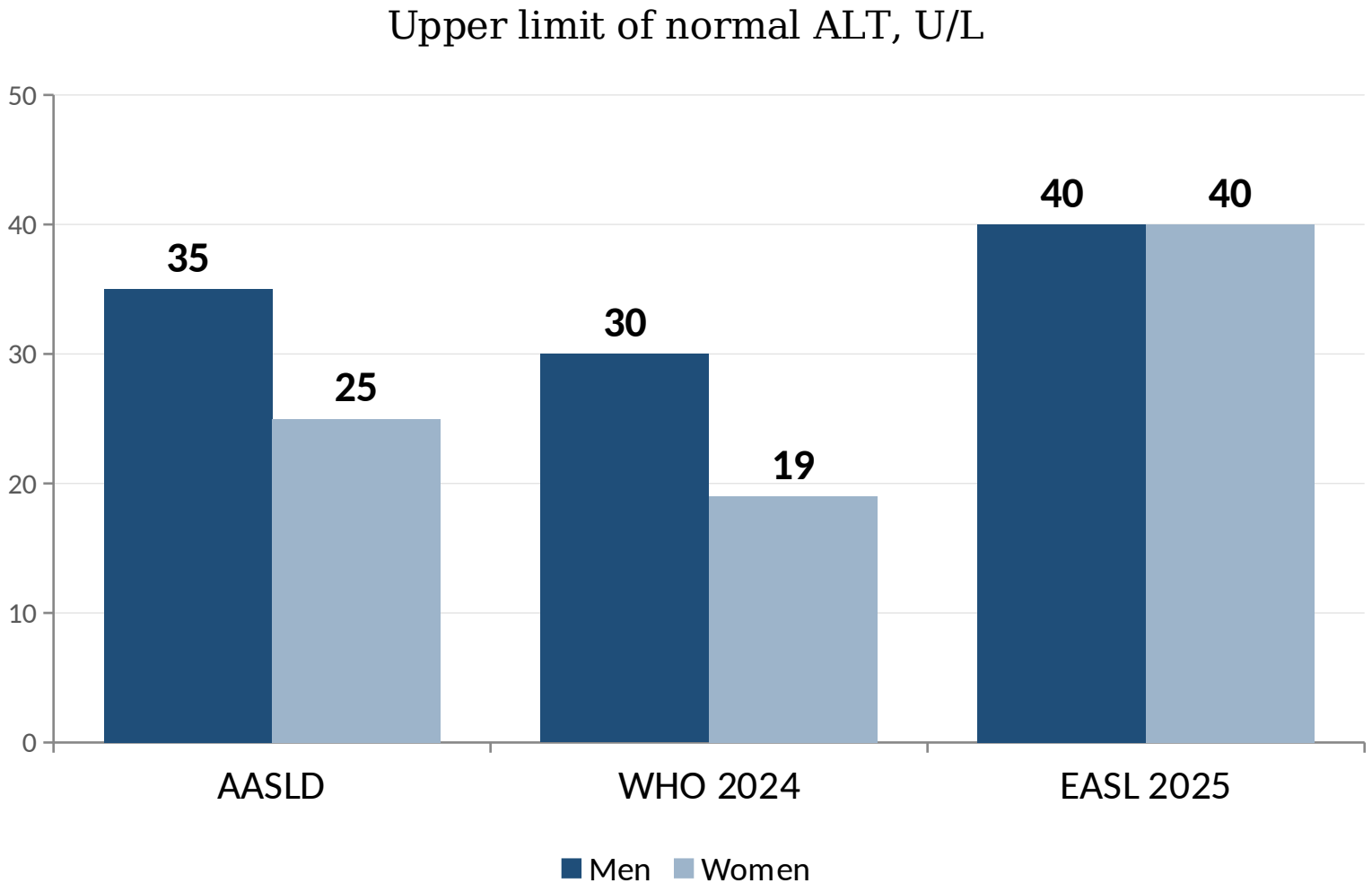
Same man, now 52 years old

Test	Result
HBsAg	Positive
HBeAg	Negative
Anti-HBe	Positive
HBV DNA	8,000 IU/mL
ALT	48 U/L
Platelets	190 ×10⁹/L
FIB-4	1.8
Liver stiffness	6.8 kPa
eGFR	45 mL/min



Do you treat him?

What counts as a normal ALT?



His ALT

48 U/L

Above all three limits

Upper limits of normal as listed in the guidelines. AASLD 2018 PMID 29405329; WHO 2024; EASL 2025 PMID 40348683.

One patient, four guidelines

Hands up if you would treat him

	AASLD 2018	AASLD/IDSA 2025	EASL 2025	WHO 2024
Age 34, HBeAg positive				
Age 41, same labs				
Age 52, HBeAg negative				

AASLD 2018 PMID 29405329; AASLD/IDSA 2025 Hepatology 2026;83:974-997 PMID 41186418; EASL 2025 PMID 40348683; WHO 2024 (NBK614987).

One patient, four guidelines

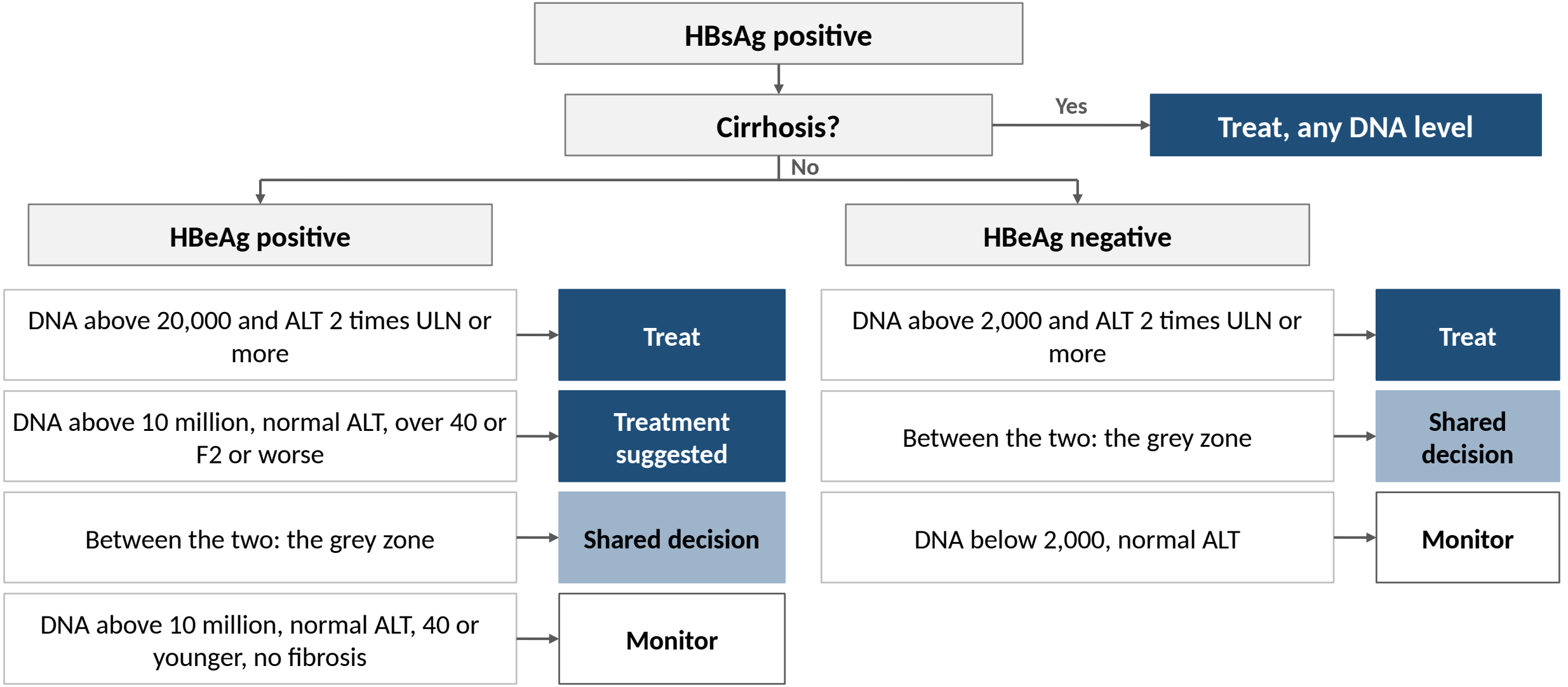
Hands up if you would treat him

	AASLD 2018	AASLD/IDSA 2025	EASL 2025	WHO 2024
Age 34, HBeAg positive	Monitor	Monitor	Consider treating	Monitor
Age 41, same labs	Monitor	Treatment suggested	Consider treating	Monitor
Age 52, HBeAg negative	Check fibrosis	Shared decision	**Treat**	**Treat**

At 52 he chose to start treatment

AASLD 2018 PMID 29405329; AASLD/IDSA 2025 Hepatology 2026;83:974-997 PMID 41186418; EASL 2025 PMID 40348683; WHO 2024 (NBK614987).

Who to treat: summary (US)



AASLD 2018 (PMID 29405329); AASLD/IDSA 2025 R3, R4 and Figure 5 (PMID 41186418). ALT ULN 35 men, 25 women.

Same man at 52, eGFR 45

Which nucleos(t)ide analogue for him?

	Dose	At eGFR 30 to 49	
Entecavir	0.5 mg daily		
TDF	300 mg daily		
TAF	25 mg daily with food		

US prescribing information: Baraclude, Viread, Vemlidy (DailyMed). Renal doses use creatinine clearance.

Same man at 52, eGFR 45

Which nucleos(t)ide analogue for him?

	Dose	At eGFR 30 to 49	
Entecavir	0.5 mg daily	0.5 mg every 48 hours	Dose-adjust
TDF	300 mg daily	300 mg every 48 hours	Dose-adjust, kidney and bone effects
TAF	25 mg daily with food	25 mg daily	**No adjustment**

TAF, or entecavir with the dose adjusted

US prescribing information: Baraclude, Viread, Vemlidy (DailyMed). Renal doses use creatinine clearance.

Choosing the drug: summary

Situation	Drug
Most patients	Entecavir, TDF or TAF
Kidney or bone disease, or at risk	TAF, or entecavir with the dose adjusted
Prior lamivudine resistance	TDF or TAF, not entecavir
HIV	TDF or TAF as part of ART
Decompensated cirrhosis	Entecavir 1 mg or TDF (no TAF in Child-Pugh B or C)
Wants a finite course	Pegylated interferon, selected patients
On treatment	DNA every 3 months, then every 3 to 6 months

AASLD 2018 (PMID 29405329); AASLD/IDSA 2025 (PMID 41186418); US prescribing information.

Calculating PAGE-B

Variable	Category	Points
Age, years	16 to 29	0
	30 to 39	2
	40 to 49	4
	50 to 59	6
	60 to 69	8
	70 or older	10
Sex	Female	0
	Male	6
Platelets, $\times 10^9$/L	200 or more	0
	100 to 199	6
	Under 100	9

52-year-old man

Started on TAF

Platelets 190

What is his PAGE-B score?

Points as printed in EASL 2025 Table 13. PAGE-B: Papatheodoridis GV et al. J Hepatol 2016.

Calculating PAGE-B

Variable	Category	Points
Age, years	16 to 29	0
	30 to 39	2
	40 to 49	4
	50 to 59	6
	60 to 69	8
	70 or older	10
Sex	Female	0
	Male	6
Platelets, ×10⁹/L	200 or more	0
	100 to 199	6
	Under 100	9

52-year-old man

Started on TAF

Platelets 190

6 + 6 + 6 = 18

High risk: 18 or more

5-year HCC about 16 to 17%

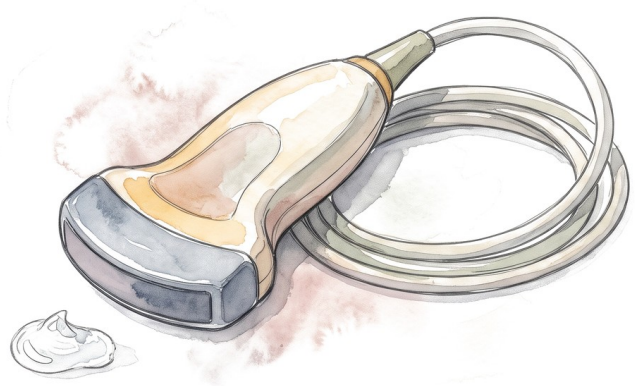
Calculator:
qaddodi.github.io/chb-hcc-risk

Points as printed in EASL 2025 Table 13. PAGE-B: Papatheodoridis GV et al. J Hepatol 2016.

Who gets surveillance?

Ultrasound and AFP every 6 months?

Patient	Surveillance
Compensated cirrhosis, Child-Pugh A	
Man from an endemic region, age 38, no cirrhosis	
Man from an endemic region, age 46, no cirrhosis	
Woman from an endemic region, age 46, no cirrhosis	
Family history of HCC	
PAGE-B score of 12	

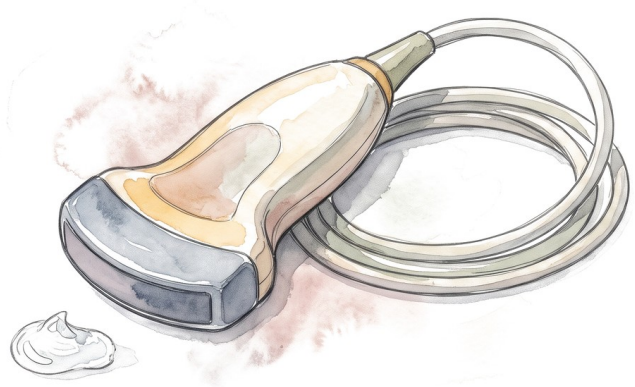


AASLD HCC practice guidance 2023, Table 1. Hepatology 2023;78:1922-1965, PMID 37199193.

Who gets surveillance?

Ultrasound and AFP every 6 months?

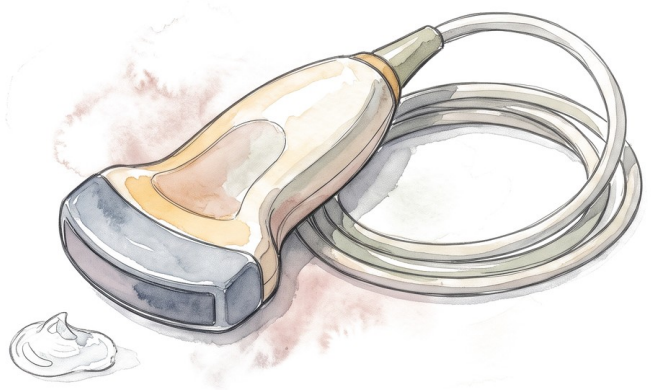
Patient	Surveillance
Compensated cirrhosis, Child-Pugh A	Yes
Man from an endemic region, age 38, no cirrhosis	No
Man from an endemic region, age 46, no cirrhosis	Yes
Woman from an endemic region, age 46, no cirrhosis	No
Family history of HCC	Yes
PAGE-B score of 12	Yes



AASLD HCC practice guidance 2023, Table 1. Hepatology 2023;78:1922-1965, PMID 37199193.

HCC surveillance: summary

- **Ultrasound and AFP every 6 months**
- Cirrhosis
- No cirrhosis, from an endemic region: men over 40, women over 50; people of African origin earlier
- Family history of HCC
- PAGE-B 10 or more
- Hepatitis D coinfection
- **Keep going after HBsAg loss if:**
- Cirrhosis or a family history of HCC, or HBsAg lost after 40 in men and after 50 in women



AASLD HCC guidance 2023, Table 1 (PMID 37199193); AASLD/IDSA 2025 R6 to R9 (PMID 41186418).

Risk of HBV reactivation

HBsAg negative, anti-HBc positive: how risky is each drug?

Drug	Reactivation risk
Rituximab or other anti-CD20	
Anthracyclines	
JAK inhibitors	
Prednisone 10 mg or more for 4 weeks or longer	
Anti-TNF	
Immune checkpoint inhibitors	
Methotrexate, azathioprine	

AGA guideline 2025, Table 6. Gastroenterology 2025;168:267-284, PMID 39863345. High above 10%, moderate 1 to 10%, low below 1%.

Risk of HBV reactivation

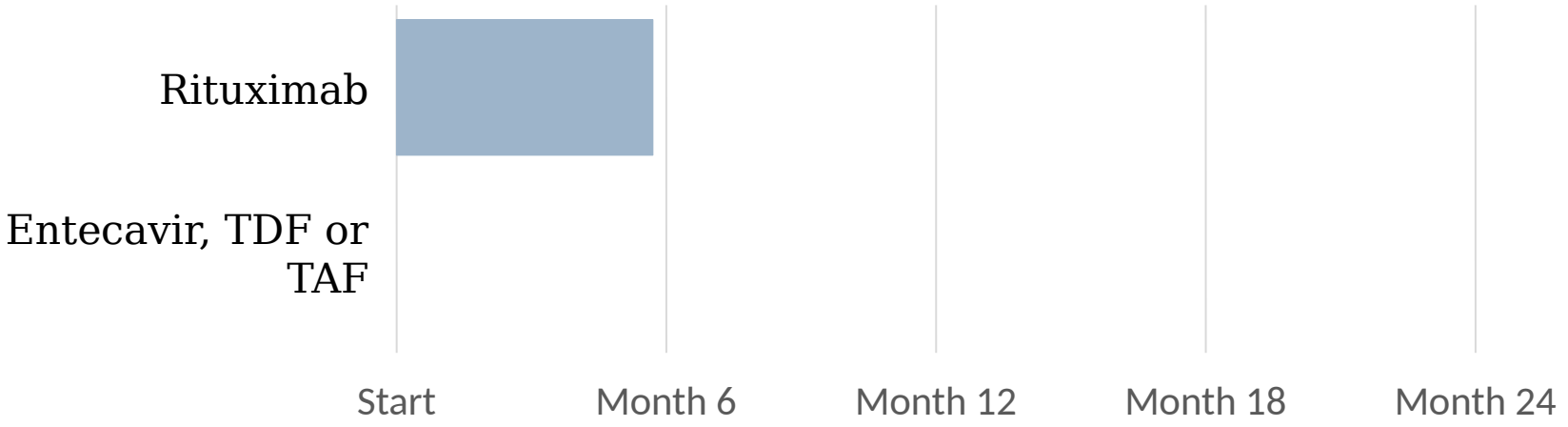
HBsAg negative, anti-HBc positive: how risky is each drug?

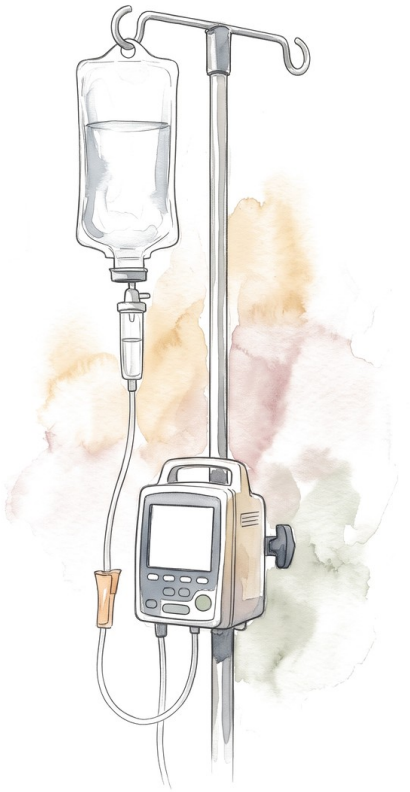
Drug	Reactivation risk
Rituximab or other anti-CD20	**High**
Anthracyclines	Moderate
JAK inhibitors	Moderate
Prednisone 10 mg or more for 4 weeks or longer	Moderate
Anti-TNF	Low
Immune checkpoint inhibitors	Low
Methotrexate, azathioprine	Low

AGA guideline 2025, Table 6. Gastroenterology 2025;168:267-284, PMID 39863345. High above 10%, moderate 1 to 10%, low below 1%.

Rituximab, six cycles

What do you give, and for how long?

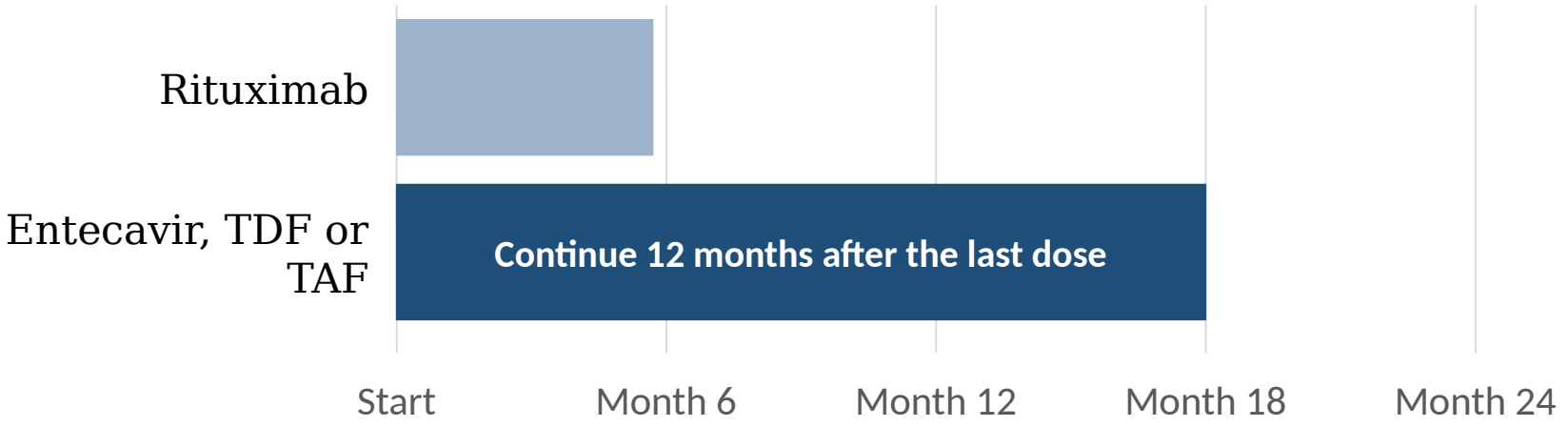


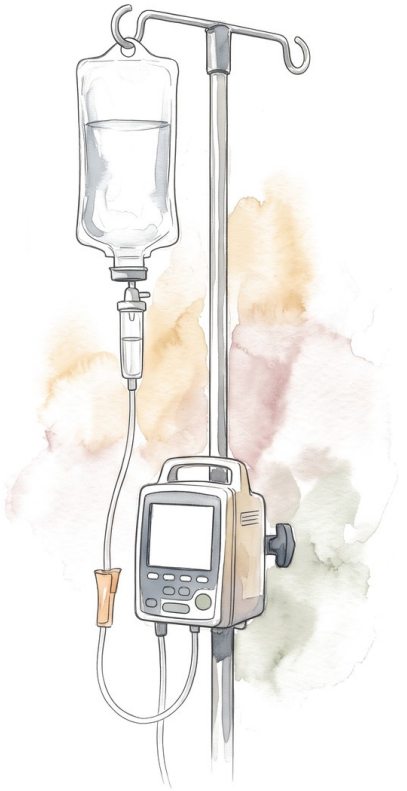


AGA 2025 (PMID 39863345); AASLD 2018 (PMID 29405329); EASL 2025 (PMID 40348683).

Rituximab, six cycles

What do you give, and for how long?

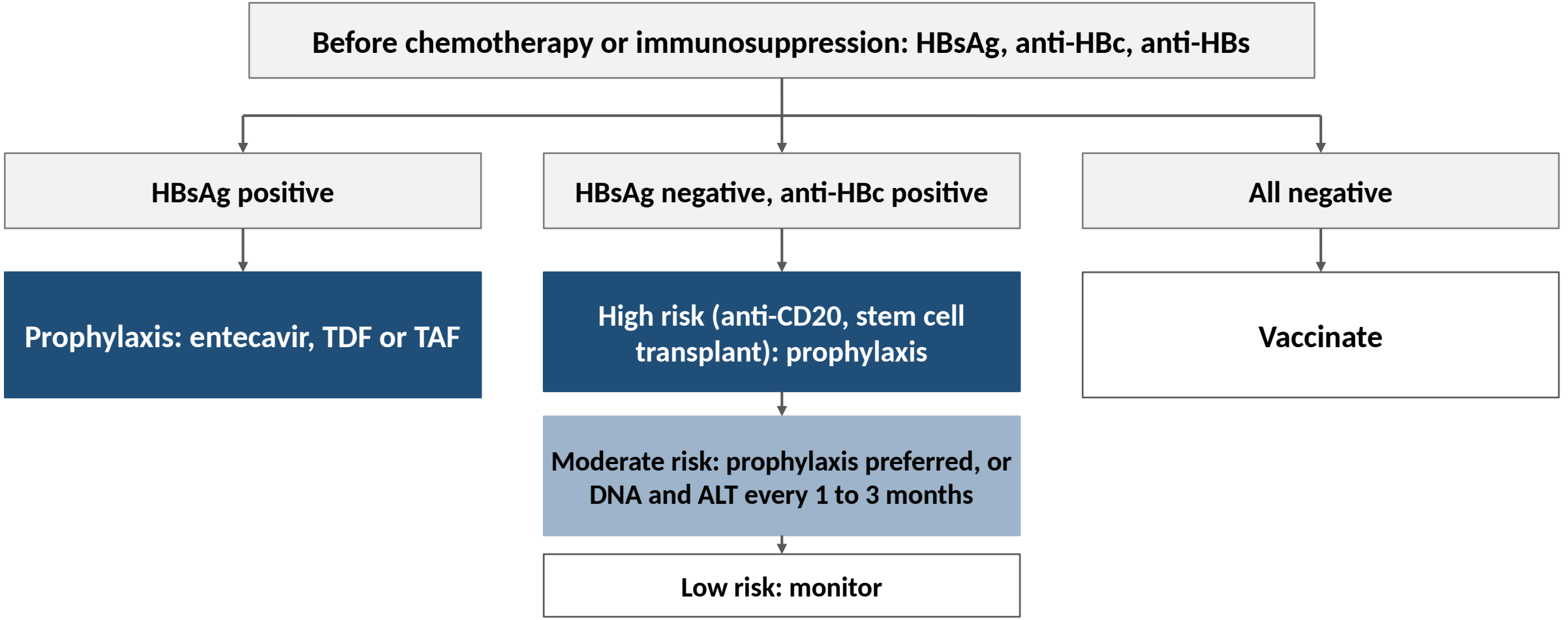




Start before the first dose

AGA 2025 (PMID 39863345); AASLD 2018 (PMID 29405329); EASL 2025 (PMID 40348683).

Before immunosuppression: summary



Prophylaxis: start before the first dose; continue 6 months after the last dose, 12 months after anti-CD20

AGA 2025 (PMID 39863345); AASLD 2018 (PMID 29405329); EASL 2025 (PMID 40348683).

29-year-old woman, 20 weeks pregnant

HBsAg	HBeAg	HBV DNA	ALT
Positive	Positive	5,000,000 IU/mL	22 U/L

What do you do for her and the baby?



AASLD/IDSA 2025 R1, Hepatology 2026;83:974-997, PMID 41186418. CDC perinatal hepatitis B guidance.

29-year-old woman, 20 weeks pregnant

HBsAg	HBeAg	HBV DNA	ALT
Positive	Positive	5,000,000 IU/mL	22 U/L

What do you do for her and the baby?

- TDF or TAF from week 28, because DNA is above 200,000 IU/mL
- Baby: hepatitis B vaccine and HBIG within 12 hours of birth



AASLD/IDSA 2025 R1, Hepatology 2026;83:974-997, PMID 41186418. CDC perinatal hepatitis B guidance.

Same woman, after delivery

She is on TDF. Can she breastfeed?

A. No, tenofovir in breast milk harms the baby

B. Yes

C. Only after the vaccine series

D. Only if her DNA is undetectable



AASLD 2018 (PMID 29405329); AASLD/IDSA 2025 (PMID 41186418).

Same woman, after delivery

She is on TDF. Can she breastfeed?

A. No, tenofovir in breast milk harms the baby

B. Yes

C. Only after the vaccine series

D. Only if her DNA is undetectable



AASLD 2018 (PMID 29405329); AASLD/IDSA 2025 (PMID 41186418).

Pregnancy: summary

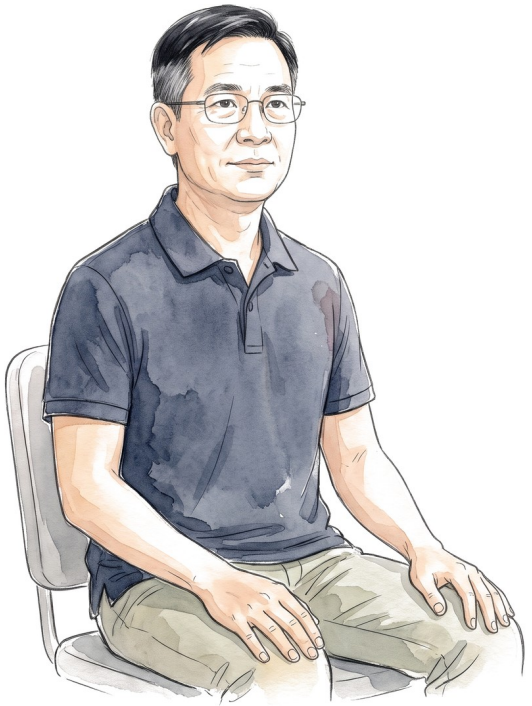
- Test HBsAg in every pregnancy
- HBsAg positive: check DNA and ALT
- **DNA above 200,000: TDF or TAF from week 28**
- Baby: hepatitis B vaccine and HBIG within 12 hours of birth, then finish the series
- Test the baby at 9 to 12 months: HBsAg and anti-HBs
- Breastfeeding is fine
- If she stops after delivery: DNA and ALT every 1 to 3 months for 6 months



AASLD/IDSA 2025 R1 (PMID 41186418); CDC 2023 (PMID 36893044); CDC perinatal hepatitis B guidance.

Same man, 57 years old, five years on TAF

Condition for stopping	Our patient	Met?
No cirrhosis or advanced disease	Stiffness 5.9 kPa	
HBV DNA undetectable for 2 years or more	Undetectable for 5 years	
HBsAg under 100 IU/mL	400 IU/mL	
HBeAg seroconversion 1 year or more	In his late 40s	
No HIV or HDV	Both negative	
Reliable follow-up	Yes	

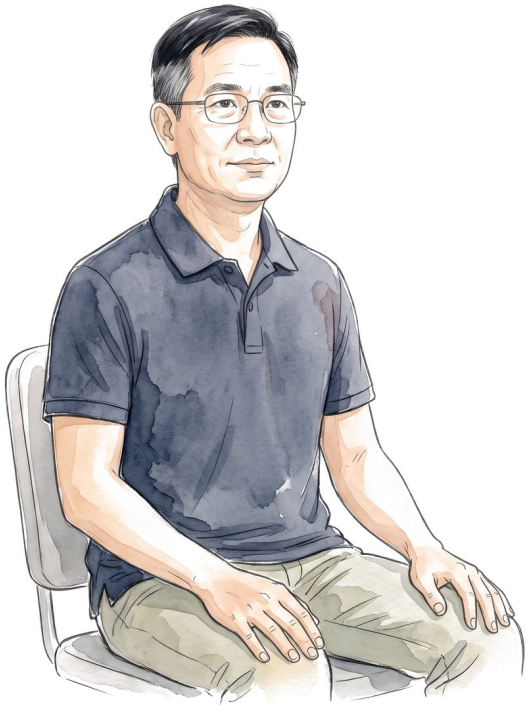


Can he stop?

AASLD/IDSA 2025 R5. Hepatology 2026;83:974-997, PMID 41186418.

Same man, 57 years old, five years on TAF

Condition for stopping	Our patient	Met?
No cirrhosis or advanced disease	Stiffness 5.9 kPa	Yes
HBV DNA undetectable for 2 years or more	Undetectable for 5 years	Yes
HBsAg under 100 IU/mL	400 IU/mL	**No**
HBeAg seroconversion 1 year or more	In his late 40s	Yes
No HIV or HDV	Both negative	Yes
Reliable follow-up	Yes	Yes



Can he stop?

No. Continue until HBsAg is lost

AASLD/IDSA 2025 R5. Hepatology 2026;83:974-997, PMID 41186418.

Same man: if he had stopped TAF

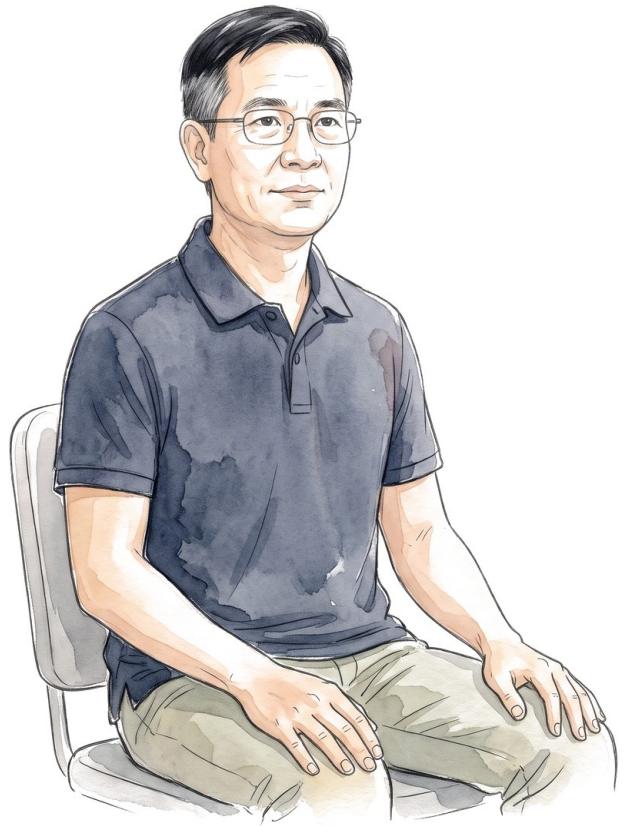
Which result means restart now?

A. ALT 1.5 times ULN

B. HBV DNA 10,000 IU/mL

C. HBsAg 200 IU/mL

D. Anti-HBs negative



AASLD/IDSA 2025 Table 3 (PMID 41186418).

Same man: if he had stopped TAF

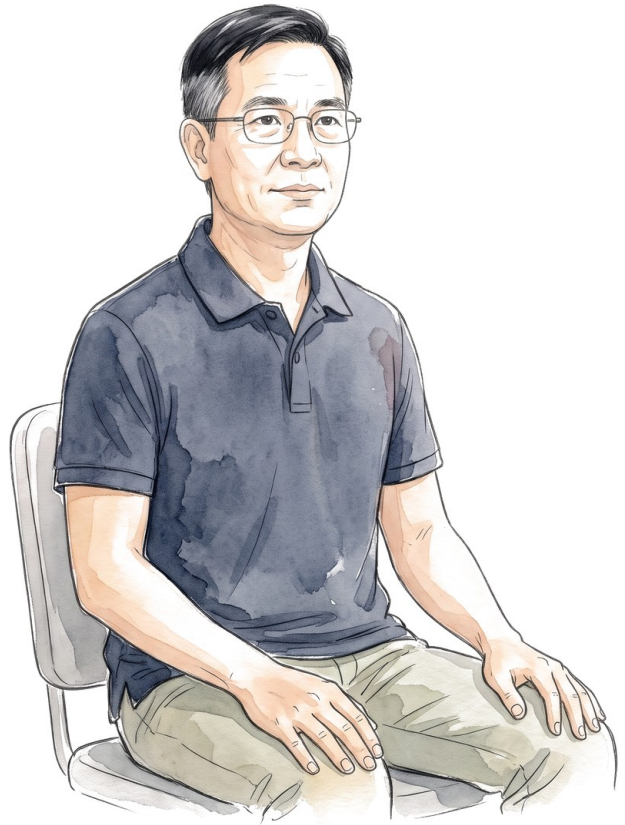
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AASLD/IDSA 2025 Table 3 (PMID 41186418).

Stopping: summary

- **Default: continue until HBsAg is lost**
- **Stop only if all of these:**
- No cirrhosis
- DNA undetectable for 2 years or more
- If HBeAg positive at the start: seroconverted 1 year or more
- HBsAg under 100 IU/mL
- No HIV or HDV
- Reliable follow-up

After stopping

Months 0 to 6: DNA and ALT every 1 to 3 months

↓

Months 6 to 12: every 3 months

↓

After 12 months: every 3 to 6 months

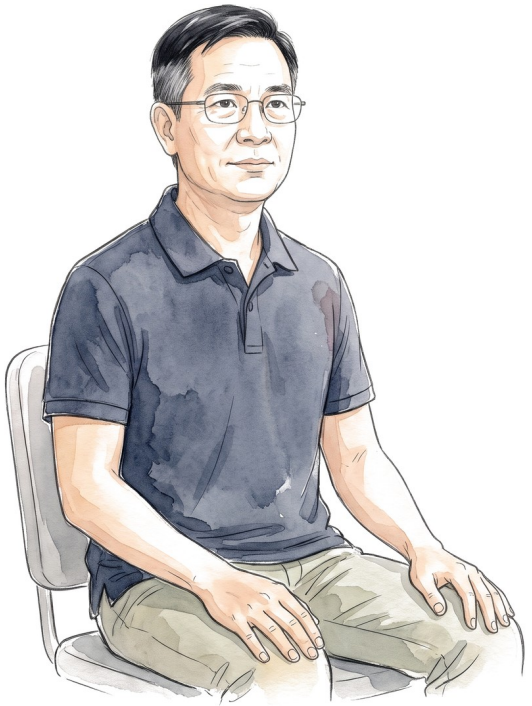
Restart now: DNA 10,000 or more, ALT above 5 times ULN, bilirubin above 2.5 mg/dL, or decompensation

AASLD/IDSA 2025 R5 and Table 3 (PMID 41186418).

Same man at 57: a new drug

Is he a candidate for bepirovirsen?

FDA label	Our patient	Met?
Adult, no cirrhosis	Stiffness 5.9 kPa	
On a nucleos(t)ide analogue 6 months or more	TAF for 5 years	
HBV DNA suppressed	Undetectable	
HBsAg 3,000 IU/mL or lower	400 IU/mL	

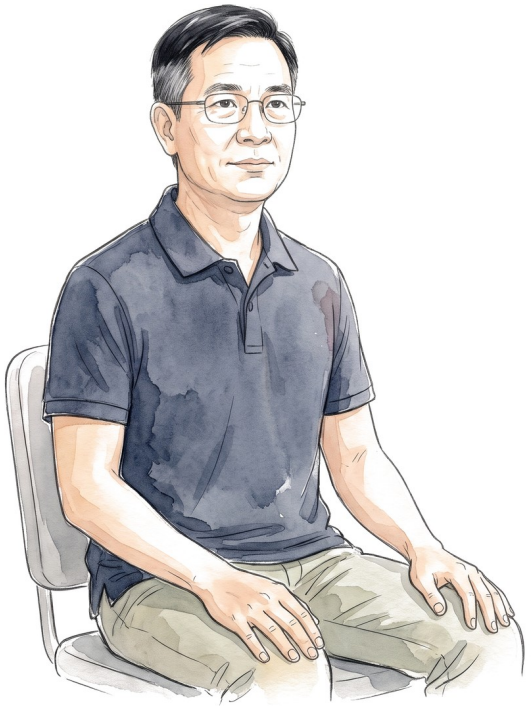


FDA news release 20 October 2026. Hou J et al. N Engl J Med 2026;394:2395-2406, PMID 42206582 (B-Well 1 and 2).

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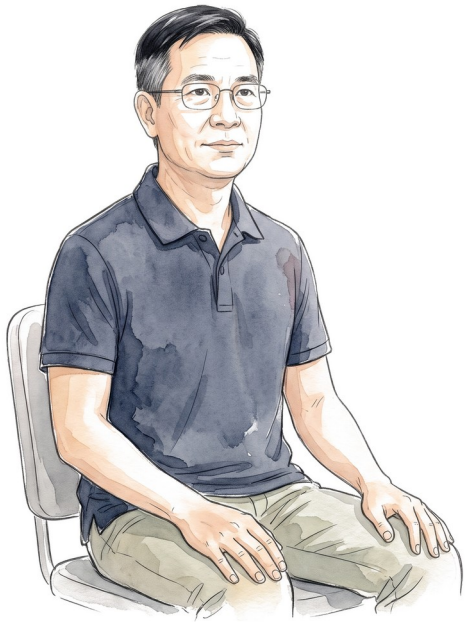


Yes. FDA approved 20 October 2026

About 1 in 5 had HBsAg loss 24 weeks after stopping all treatment; none on placebo

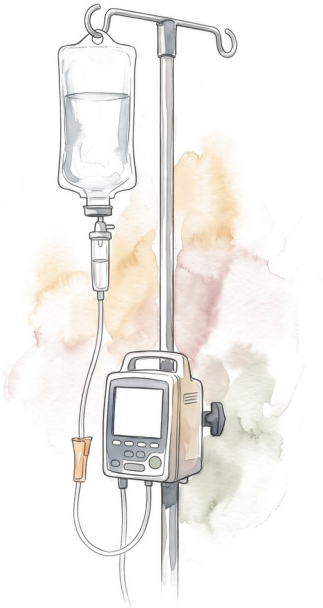
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Back to our patients



Man, 34 to 57

Monitor, surveillance from 41, TAF at 52. Now a candidate for bepirovirsen



Woman, 61, rituximab

Entecavir, TDF or TAF before the first dose, 12 months after the last



Woman, 29, pregnant

TDF or TAF from week 28. Vaccine and HBIG within 12 hours



Questions?

X: @mfalmeqdadi

Slides (PDF)